eGABAY: An AI-Powered Hybrid Public Transportation Navigation and Monitoring System for Marinduque

TEAM PIXEL6

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Abstract

Jeepneys and tricycles are essential modes of public transportation in Marinduque, serving students, workers, residents, and tourists traveling within and between barangays and municipalities. However, commuters frequently encounter difficulties accessing reliable transportation information, including departure schedules for jeepneys, terminal locations, available public transportation vehicles and its routes,and ongoing trip statuses. These challenges become more significant for tourists and individuals unfamiliar with local transportation services. Additionally, unstable internet connectivity in certain areas may limit access to digital transportation platforms, making it necessary to develop a system that remains functional even without an active internet connection.

To address these challenges, eGABAY is proposed as an AI-enhanced hybrid public transportation navigation, departure and arrival monitoring status, and information system designed specifically for Marinduque. The system will combine online and offline capabilities through a mobile application supported by a web-based administrative dashboard. It aims to provide commuters with accessible transportation information, intelligent route recommendations through ACO ant colony optimization, terminal navigation, trip monitoring, and travel assistance through a unified digital platform.

The system will incorporate an artificial intelligence-powered travel assistant capable of analyzing available transportation routes, registered terminals, destinations, schedules, jeepneys, and trip information. Users will be able to enter their desired destinations or ask transportation-related questions using natural language. Based on available data, the AI assistant will suggest appropriate jeepney or tricycle routes, estimated travel times, and relevant terminal information. The recommendations will help commuters identify efficient

transportation options while assisting tourists unfamiliar with the province's local transportation network.

As a hybrid application, eGABAY will support both online and offline functionality. In online mode, users will have access to updated trip statuses, available vehicle information, terminal directories, AI-powered travel assistance, and route recommendations. Authorized drivers will be responsible for updating their assigned trip information, including vehicle availability, route, departure point, destination, expected departure time, estimated arrival time, and trip status, such as Available, Boarding, On the Way, and Arrived/Completed. Meanwhile, the terminal personnel will be able to manage and update terminal-related information and coordinate departure schedules and vehicle availability within their respective terminals or associations. When internet connectivity is unavailable, the application will continue providing essential travel information through locally stored transportation routes, terminal directories, previously synchronized schedules, cached maps, and basic offline route guidance a local ai will also be used. Once connectivity is restored, locally stored information and permitted offline changes will synchronize with the central database. Live updates and cloud-based AI assistance will require an internet connection, while offline guidance will use locally stored route data.

Furthermore, the system will include a centralized administrative dashboard for authorized local government units (LGUs), transportation associations, and designated terminal personnel, location of terminals. This dashboard will provide organized information regarding registered vehicles, active transportation trips, terminal locations. The collected information may support transportation monitoring, route and operational planning, and informed decision-making by providing stakeholders with organized and accessible transportation data.

eGABAY aims to improve public transportation accessibility, reduce commuting uncertainty, support informed travel decisions, and enhance the overall commuting experience in Marinduque. By integrating agentic artificial intelligence, hybrid mobile application, offline accessibility, and centralized transportation information management, the proposed system seeks to bridge the information gap between commuters and transportation providers while promoting a more connected, accessible, and commuter-centered local transportation experience.